

# Yifei Zhang

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Last updated: 2026-05-10

## EDUCATION

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| <b>Ph.D. in Computer Science, Emory University</b>                                                              | Aug 2022 – Mar 2026 |
| Advisor: <a href="#">Prof. Liang Zhao</a>                                                                       |                     |
| Dissertation: <a href="#">Benchmarking, Correcting, and Accelerating AI Reasoning across Diverse Modalities</a> |                     |
| <b>M.S. in Data Science, Columbia University</b>                                                                | Sep 2020 – Dec 2021 |
| <b>Master of Engineering Management, Duke University</b>                                                        | Aug 2019 – May 2020 |
| <b>B.E. in Engineering Mechanics, Dalian University of Technology</b>                                           | Sep 2015 – Jun 2019 |

## RESEARCH INTERESTS

**Explainable AI; AI Agents and Agentic Workflows; Large Language Models; Multimodal Learning and Reasoning; Model Distillation; Evaluation and Benchmarking**

## SELECTED PUBLICATIONS

- [arXiv 2026] **Yifei Zhang**, Mingyang Li, Henry Gao, and Liang Zhao. [EmoLLM: Appraisal-Grounded Cognitive-Emotional Co-Reasoning in Large Language Models](#).
- [ICLR Workshop 2026] **Yifei Zhang**, Hooshang Nayyeri, Rinat Khaziev, Emine Yilmaz, Gokhan Tur, Dilek Hakkani-Tür, and Hari Thadakamalla. [ATOD: An Evaluation Framework and Benchmark for Agentic Task-Oriented Dialogue Systems](#). ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems.
- [arXiv 2025] **Yifei Zhang**, Jiashuo Zhang, Mojtaba Safari, Xiaofeng Yang, and Liang Zhao. [Explainable Cross-Disease Reasoning for Cardiovascular Risk Assessment from LDCT](#).
- [KDD 2025] **Yifei Zhang**, James Song, Siyi Gu, Tianxu Jiang, Bo Pan, Guangji Bai, and Liang Zhao. [Saliency-Bench: A Comprehensive Benchmark for Evaluating Visual Explanations](#). The 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Datasets and Benchmarks Track.
- [ACL Findings 2024] **Yifei Zhang**, Bo Pan, Chen Ling, Yuntong Hu, and Liang Zhao. [ELAD: Explanation-Guided Large Language Models Active Distillation](#). Findings of the 62nd Annual Meeting of the Association for Computational Linguistics.
- [IJCAI 2024] **Yifei Zhang**, Bo Pan, Siyi Gu, Guangji Bai, Meikang Qiu, Xiaofeng Yang, and Liang Zhao. [Visual Attention Prompted Prediction and Learning](#). International Joint Conference on Artificial Intelligence.
- [ICCV 2023] **Yifei Zhang**, Siyi Gu, Yuyang Gao, Bo Pan, Xiaofeng Yang, and Liang Zhao. [MAGI: Multi-Annotated Explanation-Guided Learning](#). The IEEE/CVF International Conference on Computer Vision.

Full publication list available at [Google Scholar](#).

## RESEARCH EXPERIENCE

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| <b>Doctoral Research, Emory University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Atlanta, GA         |
| <i>Explainable AI, multimodal reasoning, and efficient LLM systems</i>                                                                                                                                                                                                                                                                                                                                                                                                                         | Aug 2022 – Mar 2026 |
| <ul style="list-style-type: none"><li>– Developed benchmarks, correction methods, and acceleration strategies for AI reasoning across language, vision, and multimodal settings.</li><li>– Built explainable and multimodal learning methods for visual explanation, multimodal prediction, and agentic medical reasoning.</li><li>– Designed evaluation pipelines for LLMs and agents, including dialogue benchmarks, multimodal reasoning tasks, and reliability-focused analysis.</li></ul> |                     |
| <b>Emotionally and Reasoning-Aware LLMs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Atlanta, GA         |
| Advisor: <i>Prof. Liang Zhao, Emory University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mar 2024 – Mar 2026 |
| <ul style="list-style-type: none"><li>– Developed training and evaluation pipelines for emotionally aligned dialogue systems with structured reasoning and response planning.</li><li>– Implemented supervised fine-tuning and reinforcement learning pipelines with reward signals for reasoning quality, affective alignment, and usefulness.</li></ul>                                                                                                                                      |                     |
| <b>Agentic Multimodal Medical Reasoning from LDCT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          | Atlanta, GA         |
| Advisor: <i>Prof. Liang Zhao, Emory University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                             | Nov 2024 – Mar 2026 |
| <ul style="list-style-type: none"><li>– Designed an agentic framework combining pulmonary perception, intermediate clinical reasoning, and downstream cardiovascular risk assessment from LDCT.</li><li>– Integrated multimodal evidence and cross-disease reasoning to improve both predictive accuracy and clinical interpretability over image-only baselines.</li></ul>                                                                                                                    |                     |

## INDUSTRY EXPERIENCE

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### Amazon Alexa AI

*Applied Scientist*

Boston, MA

*Apr 2026 – Present*

- Work on LLMs, AI agents, and evaluation methods for task-oriented and conversational AI systems.

### Amazon Alexa AI

*Applied Scientist Intern*

Boston, MA

*May 2025 – Aug 2025*

- Developed an evaluation framework for task-oriented dialogue systems with agentic workflows, memory-augmented architectures, and multi-turn reasoning.
- Designed synthetic dialogue generation pipelines to model complex goal dependencies, concurrency, and long-horizon contexts for benchmarking LLM-based conversational agents.

### Amazon AGI

*Research Scientist Intern*

Boston, MA

*Jun 2024 – Sep 2024*

- Explored reward modeling and preference-based evaluation to improve alignment of LLM-driven smart speaker interactions.
- Developed pipelines for synthetic preference data generation and parameter-efficient fine-tuning to improve scalable benchmarking and deployment.

## ACADEMIC SERVICE

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**Journal Reviewer:** ACM TIST, IEEE Transactions on Big Data, IEEE Transactions on Neural Networks and Learning Systems, Journal of Applied Clinical Medical Physics, Expert Systems with Applications, Information Processing and Management, Neurocomputing

**Conference Reviewer / Program Committee:** ICLR 2025–2026, CVPR 2025–2026, ICCV 2025, ECCV 2026, BMVC 2026, ACL ARR, KDD Research Track 2025–2026, KDD Datasets and Benchmarks Track 2025–2026, ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems

## TECHNICAL SKILLS

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**Programming:** Python, C/C++, MATLAB, Java, Shell

**Systems:** Linux, Git, AWS, Slurm, Docker

**Frameworks:** PyTorch, TensorFlow, Transformers, Accelerate, DeepSpeed, TRL, PEFT